

GOUTHAM BALLA

Graduate Student at State University of New York, Buffalo

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PROFESSIONAL SUMMARY

Computer Science Graduate from SUNY Buffalo with experience in developing scalable web applications using Java, Spring Boot, and PostgreSQL, with additional expertise in Python, and RESTful API design. Proficient in modern frontend technologies like React.js, HTML, CSS, and JavaScript, and experienced with cloud platforms (AWS), containerization (Docker, Kubernetes), and CI/CD tools like Jenkins. Skilled in Git, Unix/Linux environments, and software development best practices, with a strong focus on clean, maintainable code and delivering real-world impact.

EDUCATION

Master of Science in Computer Science

Jan 2024 - May 2025

University at Buffalo, The State University of New York, USA

CGPA: 3.4/4.0

PROFESSIONAL EXPERIENCE

Software Engineer Intern | Virtusa | Hyderabad, India

May 2023– Dec 2023

- Developed and maintained responsive web applications using React.js for the frontend and ASP.NET Core RESTful APIs for the backend, leveraging Microservices architecture to enhance scalability and user experience.
- Created high-performance RESTful APIs capable of processing large data volumes (1 million records in approximately 3 seconds), increasing data processing throughput by 50%.
- Developed and deployed **CI/CD pipelines** with Azure DevOps and XENA framework, reducing deployment time by **40%**.
- Engineered interactive, user-friendly dashboards in React.js, integrating with Spring Boot and third-party APIs to deliver real-time project-level insights (Xray reports).
- Programmed reusable, **object-oriented UI components** for rapid prototyping, reducing development time by 30%.
- Implemented collection and analysis of **quantitative and qualitative UX metrics**, supporting business KPIs and driving continuous improvement in user-centered designs.
- Participated in 98% of the weekly status meetings and client interactions to gather and refine requirements, ensuring alignment with business needs and project goals.

Software Engineer Intern | Sanjana Logistics LLP | Hyderabad, India

May 2022– Aug 2022

- Engaged in the development of web-applications (Java, Spring Boot, PostgreSQL, REST API, HTML, CSS, JavaScript, React.js) in Agile methodologies.
- Enhanced database performance by refining SQL queries, implementing indexing strategies (B-tree, composite indexes), and optimizing query plans, improving execution speed by 45%.
- Leveraged **Spring Framework** enforced dependency injection, Action classes through Application Context **XML** and initialized managed beans and services.
- Collaborated in an Agile/Scrum development environment, actively participating in sprint planning, stand-up meetings, and retrospectives to drive continuous improvement.

ACADEMIC PROJECTS

Share A Ride | Full Stack Web Application

- Designed a responsive, single-page carpooling web application using **React.js**, **Node.js**, **MongoDB**, enabling users to offer, find, and manage rides based on geolocation and user preferences.
- Programmed intuitive and reusable **React.js** components for ride listings, booking forms, and user dashboards, improving user engagement and reducing page load times by **30%**.

Drowsiness Detection System | Deep Learning, Computer Vision

- Developed a real-time drowsiness detection system using a webcam to monitor eye status and trigger alerts, achieving **95.86%** test accuracy in distinguishing open/closed eye states using a **MobileNetV2**-based classifier.
- Implemented a lightweight CNN architecture optimized for real-time inference, achieving **92.8%** accuracy in drowsy-eye detection.
- Deployed **Haar Cascade** classifiers for face/eye detection and a drowsiness scoring mechanism, reducing false alarms by **15%** compared to baseline models.

LLM-Powered SQL DB Agent | LangChain, OpenAI, Python, SQL, SQLAlchemy, APIs

- Developed a natural language interface over SQL databases using LangChain + OpenAI, enabling users to query data through human-like prompts with dynamic schema understanding.
- Engineered a robust backend with FastAPI and SQLAlchemy, integrating real-time SQL validation and execution, reducing query errors by 30%+ through contextual metadata injection and error handling.

ShopSmart: AI-Driven E-commerce Comparison & Recommendations | Sentiment Analysis, LLMs

- Developed AI-powered e-commerce engine using Python, Streamlit, and OpenAI APIs, reducing product research time for users by ~40% through automated comparisons.
- Implemented aspect-based sentiment analysis (~85% accuracy) and LLM-driven summarization to extract key attributes (battery life, comfort, pricing) and deliver persona-specific recommendations, reducing manual review effort by ~60%.
- Enabled real-time dataset uploads, allowing users to customize comparisons, leading to ~30% faster decision-making in pilot tests.

CodeSphere: Real-Time Collaborative Coding Platform | React, Node.js, Socket.io, Docker, Kubernetes, Nix

- Designed a cloud-based platform enabling real-time collaborative code editing with multi-user synchronization.
- Integrated Socket.io for low-latency collaboration and Docker + Kubernetes for secure, scalable code execution environments.
- Leveraged Nix for reproducible builds and sandboxed environments, ensuring reliable and consistent execution.

TECHNICAL SKILLS

- Languages:** Python, C, Java, HTML, CSS, JavaScript, TypeScript, ES6
- Web Technologies:** Angular JS, Node.js, REST API, Spring Boot, JSON, XML, jQuery, Next.js
- Databases:** MySQL, MongoDB, PostgreSQL, AWS Cloud Services (AWS EC2, AWS Lambda, AWS RDS, AWS Elastic).
- Developer Tools/Packages:** Git, GitHub, OpenCV, TensorFlow, Django, Flask, Tableau

CERTIFICATIONS

- AWS Certified Cloud Practitioner-6014e7a317bc48f09127d90cdd113d46